4) RANDOM - INITIALIZATION (OR GATE)

```
import numpy as np
x1 = np.array([0,0,1,1])
x2 = np.array([0,1,0,1])
t = np.array([0,1,1,1])
m = len(t)
max_iter = 4000
learn_rate = 0.1
w1 = np.random.uniform(-1,1)*0.2
w2 = np.random.uniform(-1,1)*0.2
b = 0
def sigmoid(z):
    return 1/(1+np.exp(-z))
def sigmoid_derivative(s):
    return s*(1-s)
for _ in range(max_iter):
    dw1 = 0
    dw2 = 0
    db = 0
    for i in range(m):
        z = x1[i]*w1 + x2[i]*w2 + b
        y = sigmoid(z)
        error = t[i] - y
        derivative_y = sigmoid_derivative(y)
        dw1 += error * derivative_y * x1[i]
        dw2 += error * derivative_y * x2[i]
        db += error * derivative_y
    w1 += learn_rate *(dw1/m)
    w2 += learn_rate * (dw2/m)
    b += learn_rate * (db/m)
print("w1:", w1, "w2:", w2, "b:", b)
final_z = x1*w1 + x2*w2 + b
prediction = sigmoid(final_z)
print("prediction:", prediction.round())
```

OUTPUT

```
w1: 3.303033690898106 4
w2: 3.31084522961079 3
b: -1.34428171104466 9
prediction: [0. 1. 1. 1.]
```